Lecture Scribe Notes

L18.pdf
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Generate a lecture scribe notes which can be used as exam reference material .Use only the provided context (lecture slides) dont add anything of you own . While writing the scribe reason step by step through all definitions , notation, assumptions , theorem statements , proofs ,derivations and worked examples exactly as given in the lecture slides. Make the logical flow explicit so that a student can follow the reasoning during exam revision . Include proofs, derivations and examples only if they appear in the provided context. Do not introduce new explanations examples or material beyond the lecture scope . Maintain a formal and academic tone and strict faithfulness to the lecture structure

Below are **lecture scribe notes strictly derived from the provided lecture slides**. The structure, definitions, formulas, derivations, and examples follow the **exact logical flow of the slides without adding external explanations or material**. All content is based only on the provided lecture document.

CSE400 – Fundamentals of Probability in Computing

Lecture 18: Marginal PMF, Correlation, Covariance, Joint Gaussian Random Variables

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1. Marginal PMF

1.1 Definition

For discrete random variables X and Y, the **joint PMF** is

$$p_{XY}(x, y) = P(X = x, Y = y)$$

The **marginal PMFs** are obtained by summing over the other variable:

$$p_X(x) = \sum_y p_{XY}(x, y)$$

$$p_Y(y) = \sum_x p_{XY}(x, y)$$

Key Idea

The marginal PMF of one variable is obtained by **summing the joint PMF over all possible values of the other variable**.

1.2 Marginal PMF from a Joint PMF Table

Row sums give $p_X(x)$, and column sums give $p_Y(y)$.

Let the joint PMF table be:

$p_{XY}(x, y)$	y_1	y_2	y_3	$p_X(x)$
x_1	p_{11}	p_{12}	p_{13}	$\sum_j p_{1j}$
x_2	p_{21}	p_{22}	p_{23}	$\sum_j p_{2j}$
x_3	p_{31}	p_{32}	p_{33}	$\sum_j p_{3j}$

Thus,

$$p_X(x_i) = \sum_j p_{ij}$$

$$p_Y(y_j) = \sum_i p_{ij}$$

Probability conditions:

$$\sum_i p_X(x_i) = 1$$

$$\sum_j p_Y(y_j) = 1$$

1.3 Example

Let the joint PMF be

$p_{XY}(x, y)$	$y = 0$	$y = 1$
$x = 0$	1/8	1/4
$x = 1$	3/8	1/4

Step 1: Compute $p_X(x)$

$$p_X(0) = \frac{1}{8} + \frac{1}{4} = \frac{3}{8}$$

$$p_X(1) = \frac{3}{8} + \frac{1}{4} = \frac{5}{8}$$

Step 2: Compute $p_Y(y)$

$$p_Y(0) = \frac{1}{8} + \frac{3}{8} = \frac{1}{2}$$

$$p_Y(1) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

Hence

$$p_X(x) = \{ \begin{matrix} 3/8 & x = 0 \\ 5/8 & x = 1 \end{matrix}$$

$$p_Y(y) = \{ \begin{matrix} 1/2 & y = 0 \\ 1/2 & y = 1 \end{matrix}$$

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2. Correlation

Definition

A basic way to measure joint behavior is through the **product moment**.

In the continuous case,

$$R_{XY} = E[XY]$$

It tells us the **average value of the product** XY.

This is often called a **raw correlation measure**.

Interpretation

$$R_{XY} = E[XY]$$

This measure:

- depends on the scale of X and Y
- is computed **about the origin**
- mixes together:
  - mean values
  - spread
  - dependence

Thus it is useful as a starting point but **not a clean measure of centered dependence**.

If

$$R_{XY} = 0$$

then the variables are **orthogonal about the origin**.

Transition

To remove the effect of the means, we **center the variables first**, which leads to **covariance**.

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3. Covariance

3.1 Definition

Covariance is defined as

$$\mathrm{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

where

$$\mu_X = E[X], \quad \mu_Y = E[Y]$$

Interpretation

- We first subtract the mean from each variable.
- Then we measure how the **centered quantities vary together**.
- Covariance captures whether deviations from the means occur:
  - in the same direction
  - in opposite directions.

Interpretation of sign:

Covariance	Meaning
$Cov(X, Y) > 0$	Positive covariance
$Cov(X, Y) < 0$	Negative covariance
$Cov(X, Y) = 0$	Zero covariance

Covariance tells the **direction of joint linear variation** between X and Y.

However, the **magnitude depends on the units of the variables**, therefore it cannot be used to assess the strength of the relationship.

3.2 Algebraic Form and Properties

Start with

$$Cov(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

Expand:

$$= E[XY - X\mu_Y - \mu_X Y + \mu_X \mu_Y]$$

Apply linearity of expectation:

$$= E[XY] - \mu_Y E[X] - \mu_X E[Y] + \mu_X \mu_Y$$

Since

$$\mu_X = E[X], \quad \mu_Y = E[Y]$$

Therefore

$$Cov(X, Y) = E[XY] - \mu_X \mu_Y$$

Thus

$$Cov(X, Y) = E[XY] - E[X]E[Y]$$

3.3 Properties of Covariance

1. **Symmetry**

$$Cov(X, Y) = Cov(Y, X)$$

2. **Variance as a Special Case**

$$Cov(X, X) = Var(X)$$

3. **Linearity with Respect to Constants**

$$Cov(aX + b, cY + d) = ac\, Cov(X, Y)$$

4. **Independence Implies Zero Covariance**

If

$$X \perp Y$$

then

$$Cov(X, Y) = 0$$

Important statement:

Zero covariance means **uncorrelated**, but **does not imply independence** in general.

3.4 Example 1

If

$$Y = -2X + 3$$

Find $Cov(X, Y)$.

Using the covariance formula:

$$Cov(X, Y) = E[XY] - E[X]E[Y]$$

Since

$$Y = -2X + 3$$

we obtain

$$XY = X(-2X + 3) = -2X^2 + 3X$$

and

$$E[Y] = E[-2X + 3] = -2E[X] + 3$$

Substitute into covariance formula:

$$Cov(X, Y) = E[-2X^2 + 3X] - E[X](-2E[X] + 3)$$
$$= -2E[X^2] + 3E[X] + 2(E[X])^2 - 3E[X]$$
$$= -2E[X^2] + 2(E[X])^2$$

Since

$$Var(X) = E[X^2] - (E[X])^2$$

we obtain

$$Cov(X, Y) = -2Var(X)$$

3.5 Pearson's Correlation Coefficient

Pearson's correlation coefficient measures the **effect of change in one variable when the other variable changes**.

The formula is

$$\rho_{XY} = \frac{Cov(X, Y)}{\sqrt{Var(X)Var(Y)}} = \frac{Cov(X, Y)}{\sigma_X \sigma_Y}$$

Properties:

- Measures **strength and direction** of linear relationship
- **Normalized version of covariance**
- **Unit-free**

Range:

$$-1 \leqslant \rho_{XY} \leqslant 1$$

Interpretation:

- $\rho_{XY} > 0 \rightarrow$ positive linear trend
- $\rho_{XY} < 0 \rightarrow$ negative linear trend
- $\rho_{XY} = 0 \rightarrow$ no linear correlation

3.6 Example 2

Joint PDF:

$$f_{XY}(x,y) = \begin{cases} \frac{xy}{96}, & 0 < x < 4,\ 1 < y < 5 \\ 0, & \text{otherwise} \end{cases}$$

Compute:

1. $E[X]$
2. $E[Y]$
3. $E[XY]$
4. $E[2X + 3Y]$
5. $Var(X)$
6. $Var(Y)$
7. $Cov(X, Y)$
8. ρ_{XY}

Step 1: $E[X]$

$$E[X] = \int_0^4 x f_X(x) dx$$

$$= \int_0^4 x \cdot \frac{x}{8} dx$$

$$= \frac{1}{8} \int_0^4 x^2 dx$$

$$= \frac{1}{8} [\frac{x^3}{3}]_0^4$$

$$= \frac{8}{3}$$

Thus

$$E[X] = \frac{8}{3}$$

Step 2: $E[Y]$

$$E[Y] = \int_1^5 y f_Y(y) dy$$

$$= \int_1^5 y \cdot \frac{y}{12} dy$$

$$= \frac{1}{12} \int_1^5 y^2 dy$$

$$= \frac{31}{9}$$

Thus

$$E[Y] = \frac{31}{9}$$

Step 3: $E[XY]$

Since X and Y are independent,

$$E[XY] = E[X]E[Y]$$

$$= \frac{8}{3} \cdot \frac{31}{9}$$

$$= \frac{248}{27}$$

Step 4: $E[2X + 3Y]$

Using linearity:

$$E[2X + 3Y] = 2E[X] + 3E[Y]$$

$$= 2\left(\frac{8}{3}\right) + 3\left(\frac{31}{9}\right)$$

$$= \frac{47}{3}$$

Step 5: $Var(X)$

$$Var(X) = E[X^2] - (E[X])^2$$

Compute

$$E[X^2] = 8$$

Thus

$$Var(X) = 8 - (\frac{8}{3})^2$$

$$= \frac{8}{9}$$

Step 6: $Var(Y)$

$$Var(Y) = E[Y^2] - (E[Y])^2$$

$$E[Y^2] = 13$$

$$Var(Y) = 13 - (\frac{31}{9})^2$$

$$= \frac{92}{81}$$

Step 7: Covariance

$$Cov(X, Y) = E[XY] - E[X]E[Y]$$

$$= \frac{248}{27} - \frac{248}{27}$$

$$= 0$$

Thus

$$Cov(X, Y) = 0$$

Step 8: Pearson Correlation

$$\rho_{XY} = \frac{Cov(X, Y)}{\sqrt{Var(X)Var(Y)}}$$

Since $Cov(X, Y) = 0$

$$\rho_{XY} = 0$$

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4. Jointly Gaussian Random Variables

Definition

A pair (X, Y) is **jointly Gaussian** if its **joint density has the bivariate Gaussian form** and the **marginals are Gaussian**.

Applications:

- signal processing
- communication systems
- modeling continuous noisy measurements

Joint Gaussian PDF

For two variables X and Y:

$$f_{XY}(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left(\frac{-1}{2(1-\rho^2)}\left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2 - 2\rho\left(\frac{x-\mu_X}{\sigma_X}\right)\left(\frac{y-\mu_Y}{\sigma_Y}\right)\right]\right.$$

Where

- μ_X, μ_Y = means
- σ_X, σ_Y = standard deviations
- ρ = Pearson correlation coefficient

Example Interpretation

Example variables:

- X: Atmospheric CO_2 concentration (ppm)
- Y: Global surface temperature anomaly (°C)

Because both variables vary due to many environmental processes (solar radiation, ocean circulation, emissions, weather variability), their joint behavior can often be approximated by a **Joint Gaussian Distribution**.

If

$$\rho > 0$$

then **high CO_2 levels tend to occur with higher temperatures**.

✅ These notes follow the **exact structure and derivations from the lecture slides** and can be used as **exam-revision scribe material**.

If you want, I can also convert this into a **2-page exam cheat sheet (very compressed formulas + steps)** which is much faster to revise before quizzes.